

Brazilian E-Commerce Public Dataset

3. Data Quality Assessment Report

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1. Assessment Scope

- Quality checks were applied to order_level (10,000 rows x 23 cols) and item_level (11,383 rows x 22 cols).
- Six check categories evaluated: schema, data types, missing values, business rules, referential integrity, and consistency.
- A reusable validation framework was built with standalone functions for each check type returning structured pass/fail results.

2. Overall Completeness

Table	Total Cells	Filled Cells	Missing Cells	Completeness
order_level	230,000	229,157	843	99.63%
item_level	250,426	249,668	758	99.70%

3. Missing Value Detail

- Missing values are concentrated in delivery timestamps and review fields -- both structurally expected for non-delivered or non-reviewed orders.

Column	Table	Missing	Missing %	Nature
order_delivered_customer_date	order_level	285	2.85%	Structural -- order not yet delivered
order_delivered_carrier_date	order_level	179	1.79%	Structural -- carrier handoff not recorded
avg_review_score	order_level	88	0.88%	Structural -- order never reviewed
n_reviews	order_level	88	0.88%	Same orders as avg_review_score
geolocation fields (5 cols)	order_level	35	0.35%	Zip code absent from geo lookup
order_approved_at	order_level	24	0.24%	Structural -- payment not approved
payment fields (4 cols)	order_level	1	0.01%	Single order with no payment record
product_category_name	item_level	151	1.33%	Product missing from source catalogue
product_name_lenght	item_level	151	1.33%	Same 151 products with no category
product_description_lenght	item_level	151	1.33%	Same 151 products with no category

4. Duplicate Analysis

- Zero fully duplicate rows found in all eight tables checked.
- Zero duplicate primary keys across order_id, customer_id, product_id, and seller_id.
- The raw reviews table contained 10 duplicate review_ids. These were not present in the sampled subset and required no action.

5. Referential Integrity

Relationship	Orphan Keys	Result
orders.customer_id -> customers.customer_id	0	PASS
order_items.order_id -> orders.order_id	0	PASS
order_items.product_id -> products.product_id	0	PASS
order_items.seller_id -> sellers.seller_id	0	PASS

6. Validity and Business Rule Checks

- Price <= 0: 0 records -- all item prices are strictly positive.
- Freight < 0: 0 records -- no negative shipping fees.
- Review score outside 1-5 range: 0 records.
- Delivered date before purchase date: 0 records -- temporal ordering is sound throughout.
- Installments = 0: corrected -- 0 records remain after replacing zero values with 1.

7. Validation Framework Results -- 10 of 10 Checks Passed

Check	Table	Status	Detail
Schema	order_level	PASS	All expected columns present
Data Types	order_level	NOTE	order_id loaded as str -- expected, no action needed
Missing (critical cols)	order_level	PASS	No missing in order_id, customer_id, order_status
Business Rules	order_level	PASS	Price, review score, delivery date checks all clear
Consistency	order_level	PASS	No duplicate order_ids; is_delivered flag aligned with date
Schema	item_level	PASS	All expected columns present
Business Rules	item_level	PASS	item_revenue = price + freight for all 11,383 rows
Referential Integrity	orders -> customers	PASS	0 orphan keys
Referential Integrity	items -> orders	PASS	0 orphan keys

8. Data Quality Risk Summary

- Low risk: Missing delivery dates are expected for non-delivered orders and are handled by binary flag columns.
- Low risk: 151 item-level category unknowns (1.33%) are insufficient to materially skew category-level revenue analysis.
- Medium risk: 0.4% repeat customer rate may undercount true repeat buyers if they re-register under new accounts per purchase.
- Low risk: Geolocation gaps (0.35%) are too small to distort any regional analysis.